

Welcome to Georeferencing with *QGIS*

Questions to think about:

- What is your name?
- What program/department are you from?
- What brought you to the workshop today?
- What do you hope to get out the workshop?

With *QGIS* someone could:

- Create beautiful maps...

and/or

- Conduct spatial analysis...

QGIS has hundreds of tools and possibilities...

This is an introduction to *QGIS* and georeferencing



Georeferencing with *QGIS*



As per the instructions when you signed up:
QGIS downloaded on your computer?



If not, please come back to the workshop another time!

Territorial Acknowledgment

We acknowledge and respect the ləkʷəŋən peoples on whose traditional territory the university stands and the Songhees, Esquimalt, and WSÁNEĆ peoples whose historical relationships with the land continue to this day.



Learning Objectives

- **Define** the basics of a **Geographic Information System** and **raster data**
- **Identify and navigate** key interface elements of *QGIS Georeferencer* tool
- **Define** key terms and processes of georeferencing
 - Target data, Ground Control Point, Transformation algorithm



Outcomes

Using *QGIS*, participants will:

- Use *QGIS Georeferencer* tool to georeference a historical map of Victoria, BC
- Use an existing open data source as target data to georeference to



What is a Geospatial Tool?

- Software/hardware typically designed for specific tasks or functions such as mapping
- Web-based mapping tools, very limited spatial analysis



Not all geospatial tools are a GIS!

What is a Geographic Information System (GIS)?

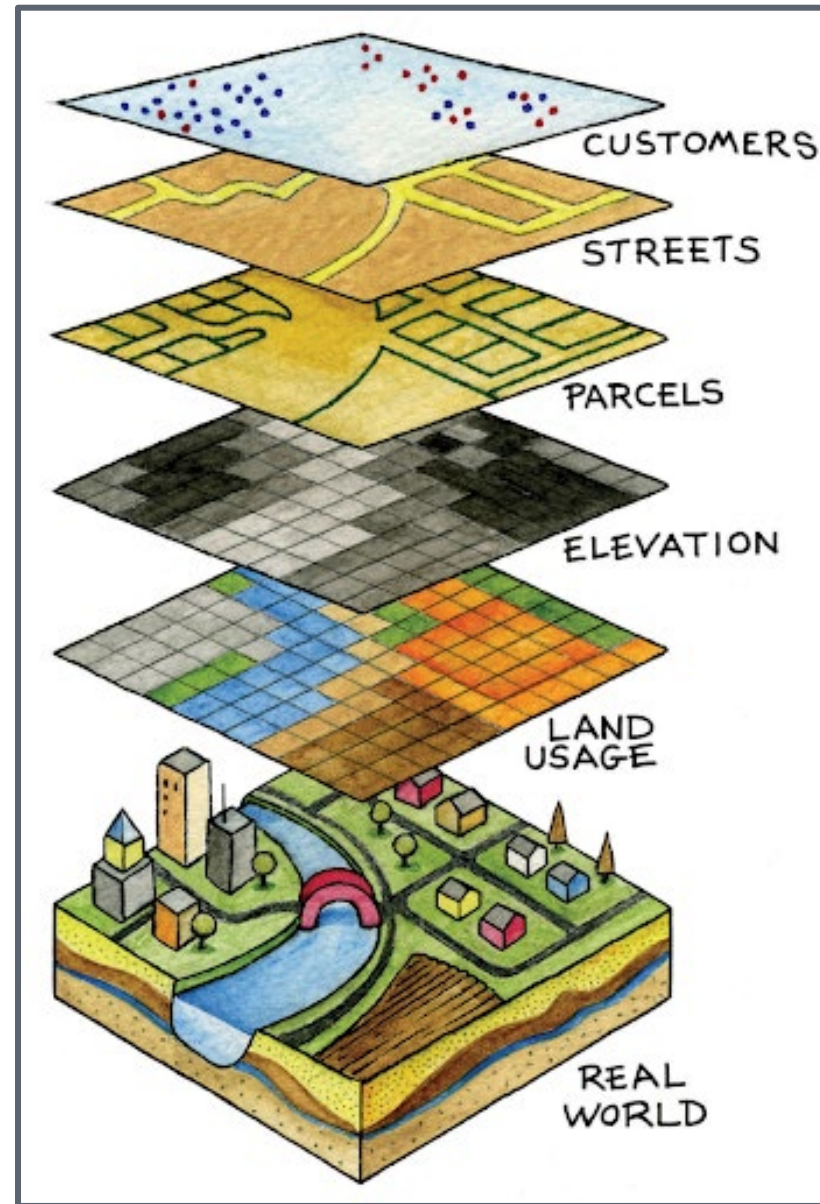
- A comprehensive system which encompasses a range of geospatial tools for spatial modelling and analysis



What is GIS?

GIS can:

- Create
 - Edit (manipulate, sub-set, etc.)
 - Query
 - Manage
 - Analyse
 - Store
- data**



GIS and Mapping - Wyoming County, PA, USA

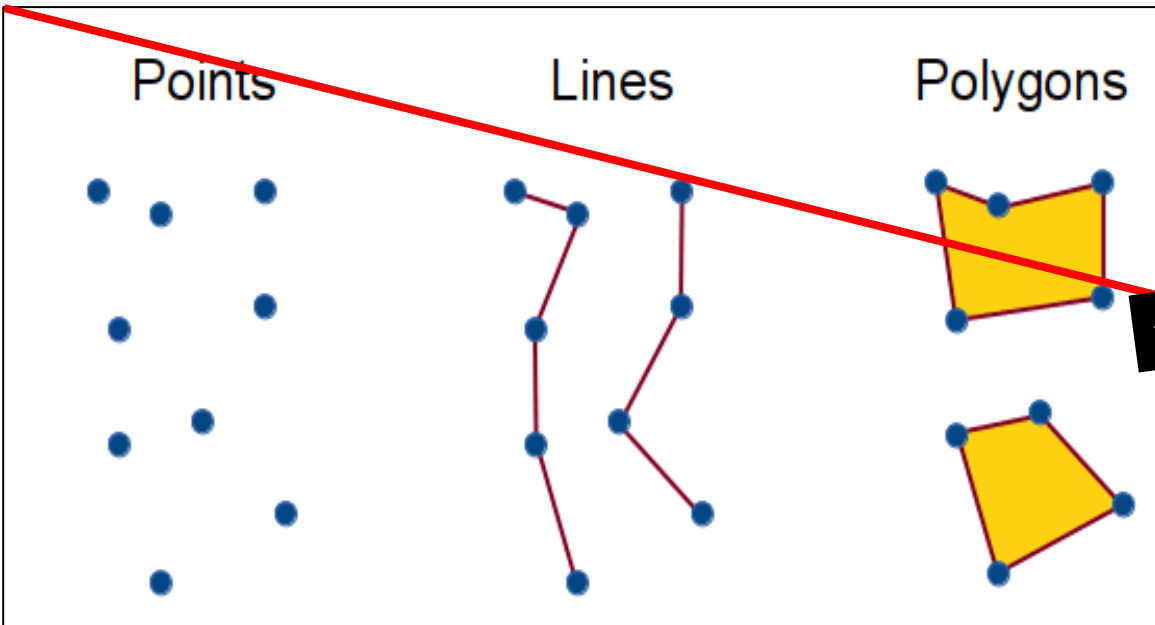
Maguire, D.J. (1991). An overview and definition of GIS.
Geographical information systems: Principles and applications.

Chrisman, N.R. (1999). What does 'GIS' mean? *Transactions in GIS* 3(2) 9

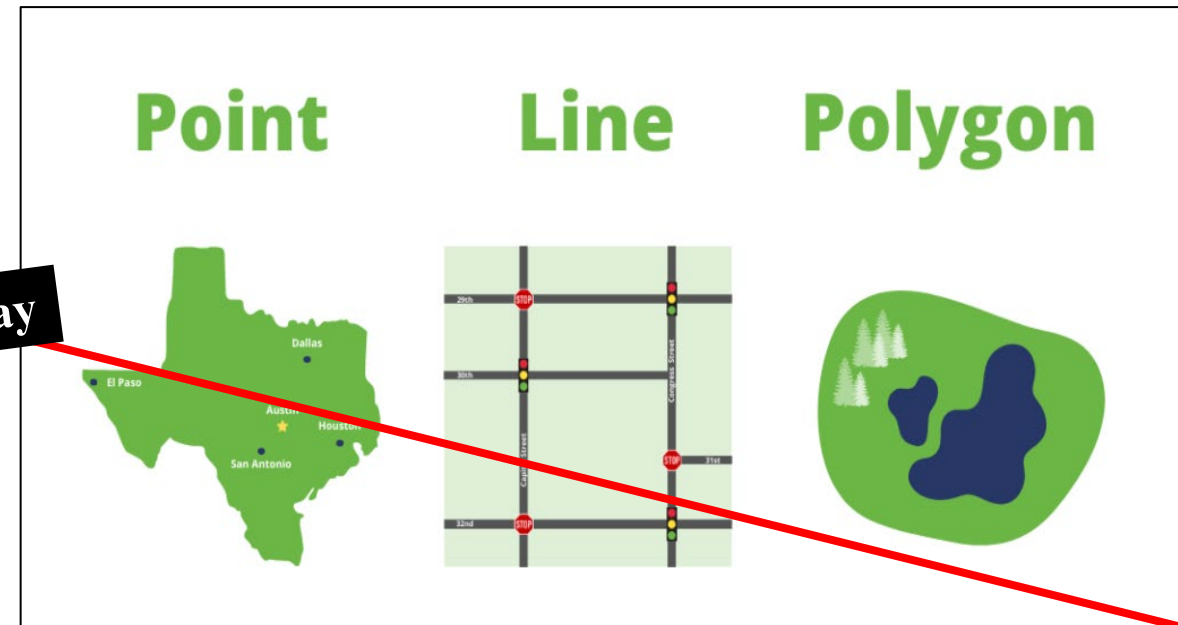
Geospatial Data: Two Types

Vector

- Sometimes no accompanying data values



[Visualizing Terrain in ArcGIS Pro](#)



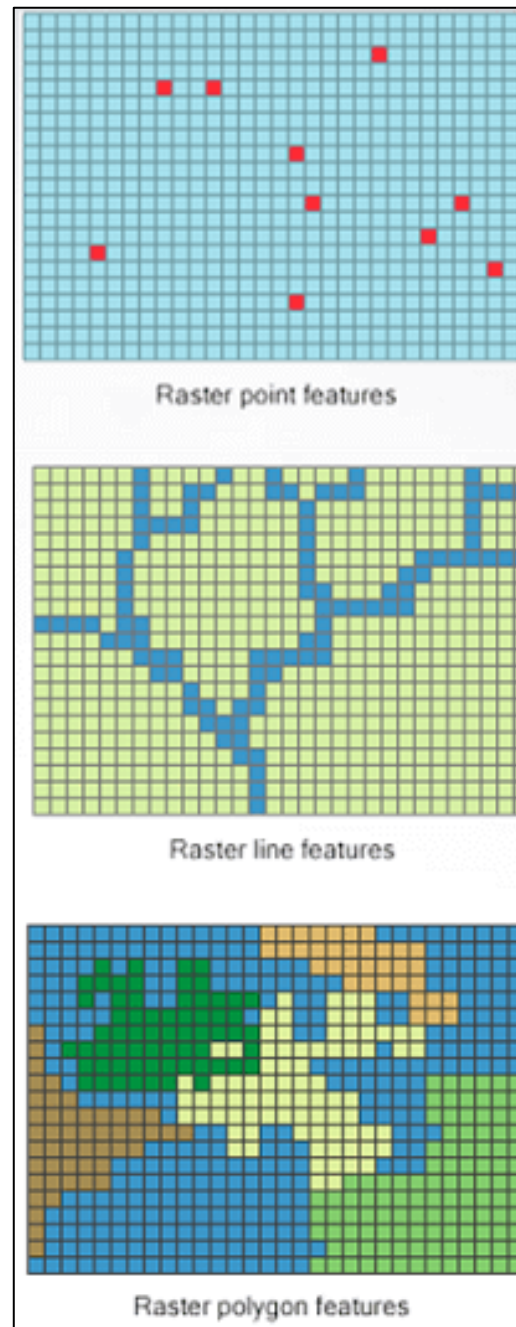
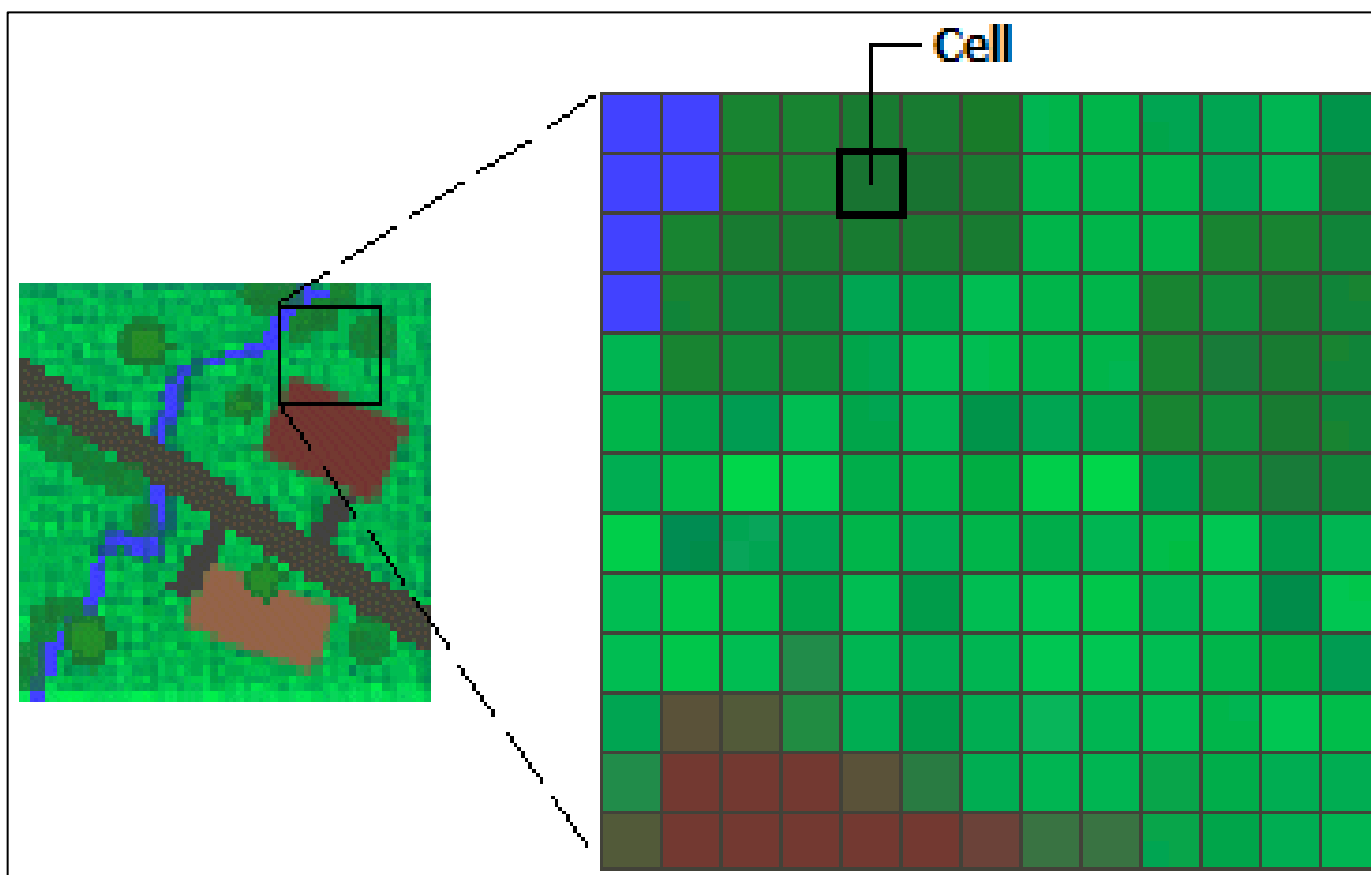
not today

[Spatial Data Models - Geographic Information Systems \(GIS\) - LibGuides at University of Connecticut](#)

Geospatial Data: Two Types

Raster

- Represents Earth's surface as a grid of pixels or cells
- Each cell has numeric value(s)



Scanned airphotos and scanned maps

- Scanned airphotos and scanned maps (sometimes historical satellite imagery) are a type of raster data
- These scanned images are made up of pixels



<https://kokoalberti.com/>

Scanned airphotos and scanned maps

- Scanned airphotos and scanned maps do not have spatial reference info
- Need to align or **georeference** it to other spatial data with a map projection



Georeferencing

To georeference scanned maps or airphotos:

1. Assign coordinate system
2. Define location using map coordinates

Can then:

- View the historical image in comparison with current

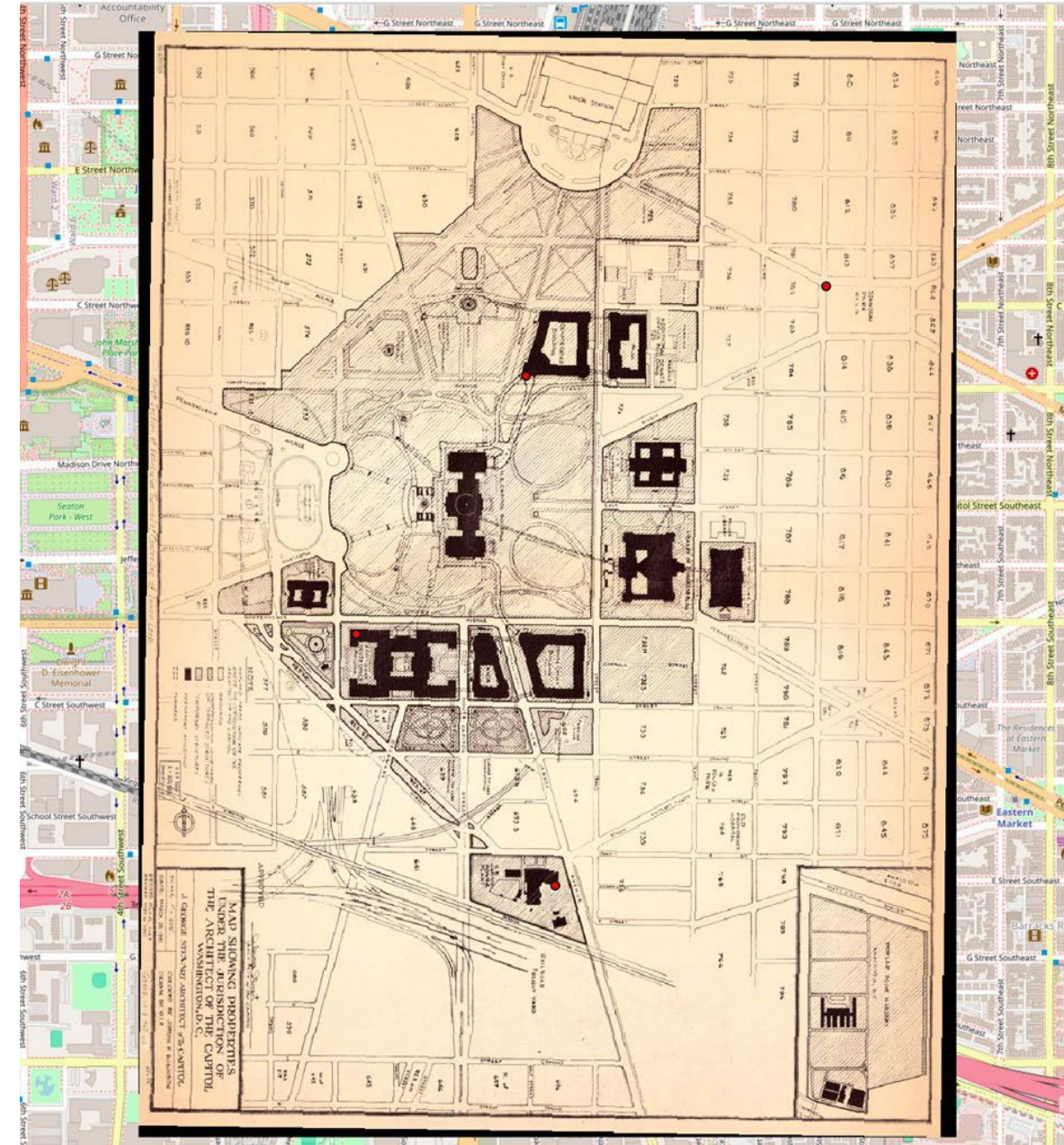
IF additional work of drawing vector polygons (for example) can:

- Analyze

and

- Query

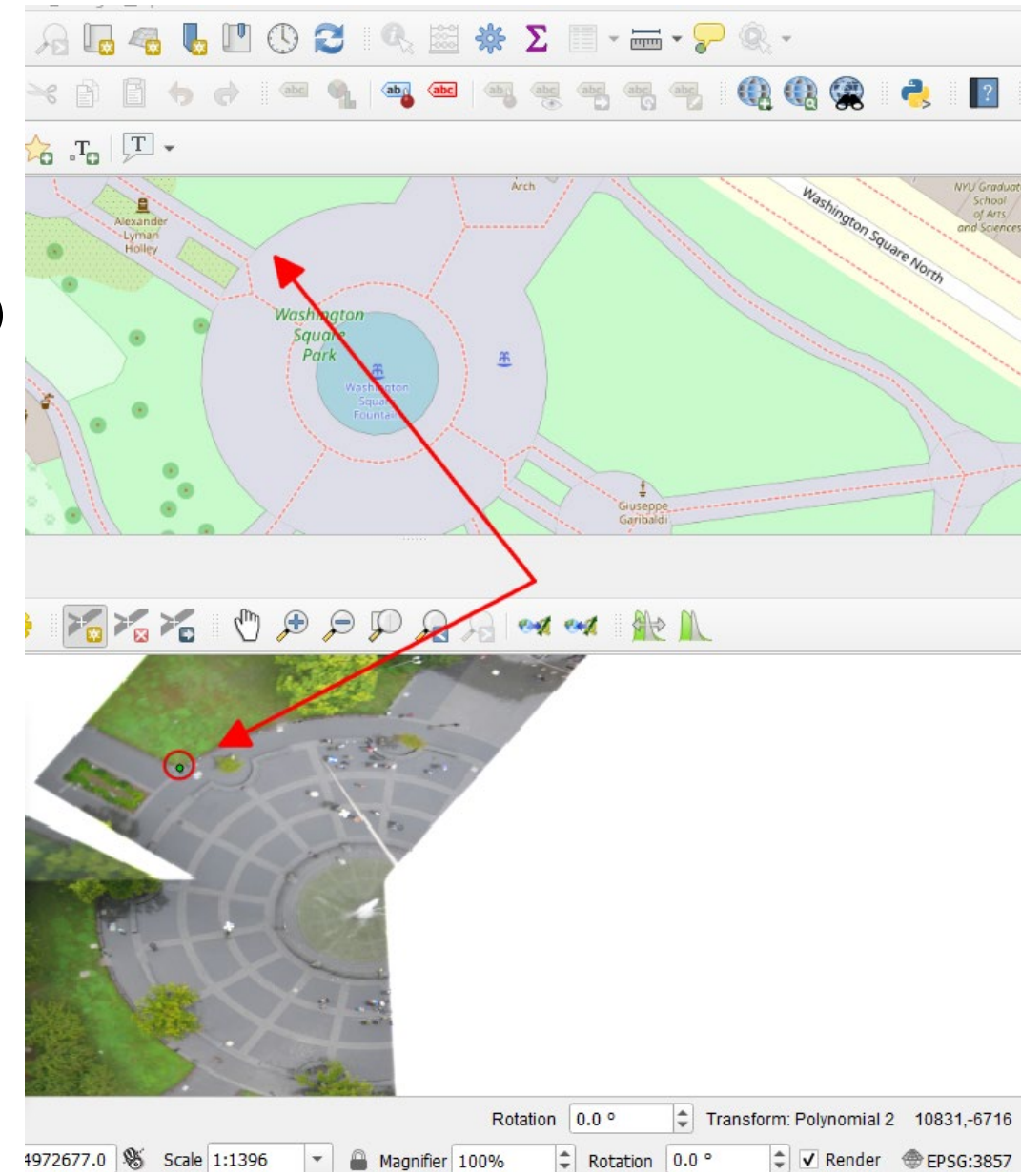
georeferenced raster with other spatial data



<https://www.geographyrealm.com/georeference-map-qgis/>

Georeferencing

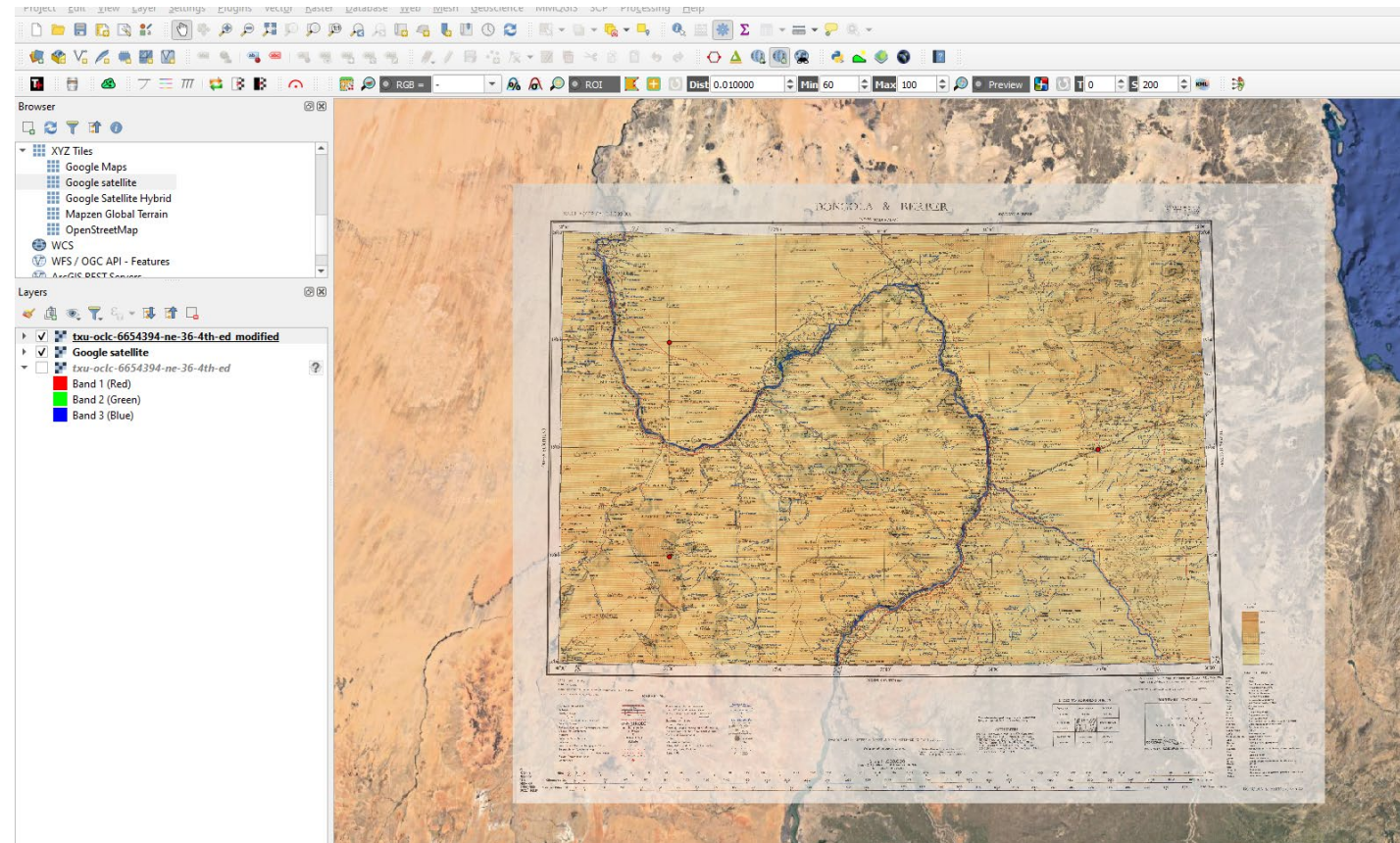
- Georeference raster using existing spatial data (**target data**) in a desired map coordinate system
- Requires using a GIS software to identify **ground control points (GCP)** with known x,y coordinates to link locations on the scanned map or airphotos with locations in the spatially referenced data (target data)



https://www.qgistutorials.com/en/docs/3/advanced_georeferencing.html

Coordinate Reference Systems

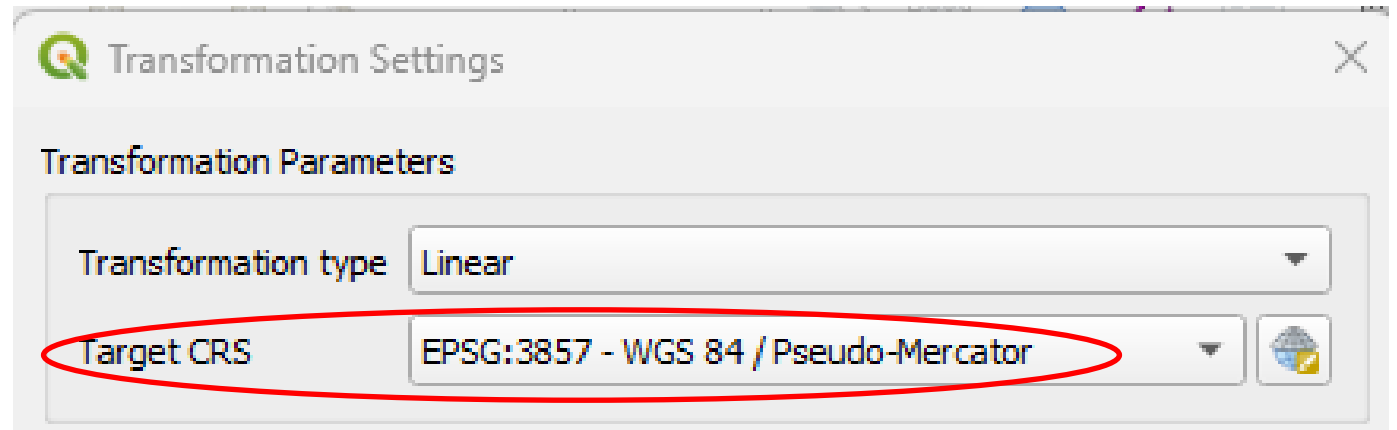
- Provides standardized framework for spatial data (latitude and longitude)
- Georeferencing = connecting raster to real-world coordinates on known coordinate system
- Raster and reference/target data need to have same coordinate system to ensure data alignment and limited distortions



<https://www.geographyrealm.com/georeference-maps-qgis/>

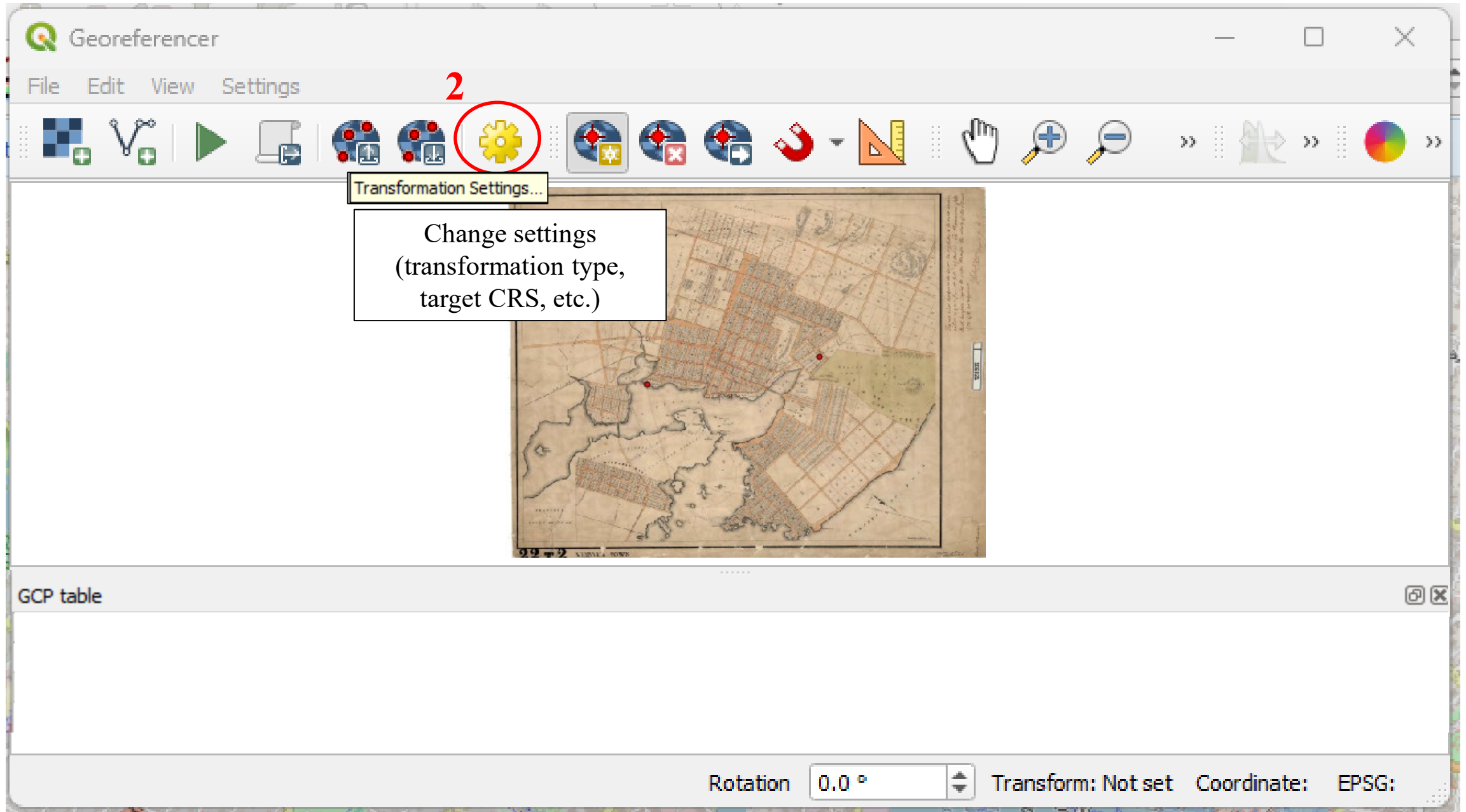
Coordinate Reference Systems

- Need *QGIS* project CRS to be the same as target CRS for georeferenced raster



EPSG:3857 is a commonly used global projection – good for georeferencing because georeferenced maps are often added to Web Mercator 3857 interfaces/platforms

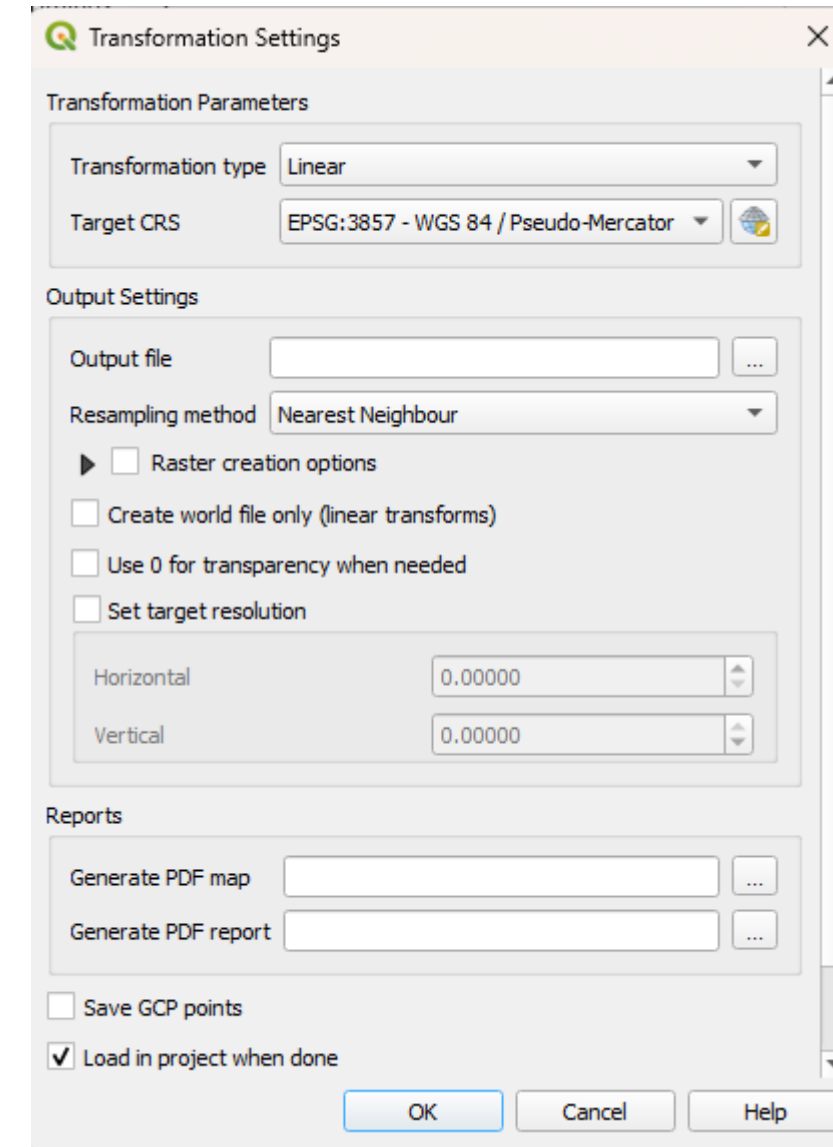
Georeferencer interface



Transformation settings

Choosing transformation type depends on:

- What you are georeferencing (type and quality of scanned airphoto or scanned map, size of area shown on map)
- How many GCPs needed
(how long are you willing to spend georeferencing?)
 - Linear: at least 2 GCPs
 - Helmert: at least 2 GCPs
 - Polynomial: at least 3 GCPs
 - Thin Plate Spline: 10+ GCPs
- We will use **Helmert**
 - Useful for good quality local map, but not aligned with grid bearing in CRS
 - Scanned map can be rotated



The screenshot shows the Georeferencer application window. At the top, there is a menu bar with 'File', 'Edit', 'View', and 'Settings'. Below the menu is a toolbar with various icons. A red circle highlights the 'Add GCP Point' icon, which is a blue circle with a white cross and a yellow star. A red number '3' is next to this icon. A text box above the toolbar says 'Add ground control point to raster then basemap'. Below the toolbar is a map of Victoria, British Columbia, with two red dots indicating ground control points. A text box below the map says 'GCP points will appear here'. At the bottom, there is a 'GCP table' with the following data:

Enabled	ID	Source X	Source Y	Dest. X	Dest. Y	dX (pixels)	dY (pixels)	Residual (pixels)
✓	0	1274.6662	-1822.9318	-13735388.97	6180446.31	0.000000	-0.000000	0.000000
✓	1	2873.5504	-1568.3323	-13732177.20	6177782.94	-0.000000	-0.000000	0.000000

At the bottom right, there is a status bar with 'Rotation 0.0 °', 'Transform: Not set', 'Coordinate: EPSG:', and a 'Close' button.

Ground control points (GCPs)

- Specific, recognizable spots on the ground with known x,y locations
 - Choose stable, easily identifiable locations (intersections, etc.)
 - As spread out on map as possible
- # of GCPs depends on:
 - time to georeference
 - size of area of map
 - how much map needs to be transformed
 - how accurately trying to match features (intersections, coastlines, etc.)

More info at:

https://docs.qgis.org/3.44/en/docs/user_manual/managing_data_source/georeferencer.html



Georeferencer interface

Once you have chosen GCPs and set transformation settings

File Edit View

Start Georeferencing

4



GCP table

Enabled	ID	Source X	Source Y	Dest. X	Dest. Y	dX (pixels)	dY (pixels)	Residual (pixels)
☒	0	1274.6662	-1822.9318	-13735388.97	6180446.31	0.000000	-0.000000	0.000000
☒	1	2873.5504	-1568.3323	-13732177.20	6177782.94	-0.000000	-0.000000	0.000000

Rotation 0.0 ° Transform: Not set Coordinate: EPSG:



Proceed to **Workshop Exercise**

There are **Check-In** slides to **ask** for help
(or **ask** sooner!)

Resources going forward:



QGIS – additional resources:

- QGIS Training Manual: https://docs.qgis.org/3.40/en/docs/training_manual/index.html
- QGIS User Guide: https://docs.qgis.org/3.40/en/docs/user_manual/index.html
- QGIS Tutorials & Tips: <https://www.qgistutorials.com/>

Find data:

- GeoSpatial Data Guide: <http://libguides.uvic.ca/geospatialdata>

Questions or problems:

- UVic Geospatial Librarian (danielbm@uvic.ca) or YCW Geospatial Intern (gabriellewade@uvic.ca)

UVic full semester GIS courses in the Department of Geography:

- GEOG222 – Intro to Maps and GIS
- GEOG328 – GIS Analysis



GIS Skills and Mapping Micro-certificate

<https://continuingstudies.uvic.ca/science-and-the-environment/programs/gis-skills-and-mapping>